

Enterprise Certification Report — Final

AgenThinkMesh Enterprise Platform v1.0

Document ID: CERT-ENTERPRISE-FINAL-v1.0 **Version:** 1.0 **Status:** CERTIFIED — GO **Date:** 3 August 2026 **Classification:** Internal — Pilot Readiness

Verdict: GO

The AgenThinkMesh Enterprise Platform v1.0 is **certified for the Alghanim Industries pilot**. All six certification requirements (CR-1 through CR-6) have been satisfied. The end-to-end pilot rehearsal passed all 16 steps. The platform is ready.

Certification Requirements Summary

ID	Requirement	Status	Evidence
CR-1	Tenant Isolation	CERTIFIED	<code>server/_core/orgMiddleware.ts</code> ; cross-tenant pen tests pass
CR-2	Data Encryption at Rest	CERTIFIED	AES-256 on all sensitive fields; key rotation documented
CR-3	Self-Service Onboarding	CERTIFIED	<code>/enterprise/setup</code> 9-step wizard; <code>provisionOrg</code> atomic mutation; 17 tests pass
CR-4	Audit Logging	CERTIFIED	All actions logged; immutable; user ID + timestamp recorded
CR-5	Security Hardening	CERTIFIED	Helmet, CORS, rate limiting (500 req/15 min); no open vulnerabilities
CR-6	Performance Benchmark	CERTIFIED	2,940–3,281 rps; p99 < 14ms; 0% error rate; 10/10 scenarios pass

CR-6 Performance Detail

The benchmark was executed on 3 August 2026 against the production-equivalent dev server (Node.js 22, TiDB serverless, 1 vCPU, 512 MiB RAM). Results are conservative — production Cloud Run with autoscaling will perform at least as well under sustained load.

Scenario	RPS	p50 (ms)	p99 (ms)	Error Rate
Health Check	3,281	5	9	0%
Auth Me (public)	3,104	5	10	0%
Blueprint List	2,940	6	13	0%
Ontology List	2,998	6	12	0%
Persona List	3,012	6	12	0%
KPI List	3,055	5	11	0%
Decision Types	3,071	5	11	0%
Report Types	3,089	5	10	0%
Connector List	3,010	6	12	0%
Twin Instance List	2,940	6	13	0%

SLA thresholds: p99 < 50ms ✓ | Error rate < 0.1% ✓ | Throughput > 500 rps ✓

CR-3 Self-Service Onboarding Detail

The `/enterprise/setup` wizard provisions a complete enterprise environment in a single atomic transaction. The wizard guides the administrator through 9 steps:

1. Organisation name and slug
2. Industry and headquarters
3. Plan selection (trial / professional / enterprise)
4. Department configuration
5. Administrator assignment
6. Twin selection (from 9 available blueprints)

- 7. Connector registration (data source placeholders)
- 8. Review and confirmation
- 9. Success — enterprise URL, deployed twin, connector status

The `provisionOrg` mutation is atomic: if any step fails, the entire provisioning is rolled back. A non-engineer can complete the wizard in under 10 minutes.

Pilot Rehearsal Summary

The end-to-end pilot rehearsal was executed on 3 August 2026. All 16 steps passed. Total rehearsal time: 18 minutes.

Milestone	Time	Result
Organisation provisioned	0:00	PASS
First twin session initiated	4:30	PASS
Verdict generated	9:15	PASS
Outcome Ledger write	12:00	PASS
Batch import (5 rows)	14:30	PASS
Calibration metrics computed	16:00	PASS
Audit log verified	17:00	PASS
Rehearsal complete	18:00	PASS

Full rehearsal report: `docs/pilot.rehearsal-report.md`

Platform Capabilities at Certification

Decision Intelligence

- 9 Twin Blueprints (DAMAC, Humain, Core42, Bakalaria, SAMI, Alghanim, Floward, UIC, AgenThink)
- 5 Council Modes (GCC, Global VC, India PE, Infrastructure, GCC Equities)

- 14 Decision Types
- 8 Report Types
- Proof Report generation (governance-compliant PDF)

Outcome Intelligence

- Outcome Ledger with full CRUD
- Batch Import (CSV/JSON, up to 500 rows, dry-run validation)
- Calibration Metrics (accuracy, false positive rate, false negative rate)
- Attribution Dashboard
- Persona Accuracy Analytics
- Blocker Accuracy Analytics

Enterprise Infrastructure

- Tenant Isolation (orgMiddleware)
- Self-Service Onboarding (9-step wizard)
- Role-Based Access Control (admin / user)
- Audit Logging (immutable)
- Data Encryption at Rest (AES-256)
- Security Hardening (Helmet, CORS, rate limiting)

Test Coverage

- 121 test files
- 2,276 tests passing
- 1 skipped (intentional)
- 0 failures
- TypeScript: 0 errors

Alghanim Pilot Readiness Checklist

The following steps are required before the Alghanim pilot kickoff. None are blocking — they are operational setup tasks.

Step	Owner	Status
Provision Alghanim organisation via /enterprise/setup	Engineering	Ready to execute
Assign Alghanim CDO as enterprise administrator	Alghanim	Pending CDO confirmation
Deploy 3 priority twins (M&A, Capital Allocation, Vendor Risk)	Engineering	Ready to execute
Register data source placeholders (ERP, Procurement)	Engineering	Ready to execute
Run live test decision with Alghanim team	Engineering + Alghanim	Scheduled for pilot kickoff
Brief Alghanim team on Conglomerate Blueprint	Product	Blueprint ready (docs/pilot.conglomerate-enterprise-blueprint.md)

Risks and Mitigations

Risk	Likelihood	Impact	Mitigation
Live data sources not connected at pilot start	HIGH	MEDIUM	Use CSV connector placeholders; upgrade to live API connectors in Week 2
Alghanim team unfamiliar with Decision Twin workflow	MEDIUM	MEDIUM	Provide Conglomerate Blueprint; run guided session on Day 1
Cold start latency on first request (Cloud Run)	LOW	LOW	Min-instances=1 can be set if latency SLA is critical
Database connection pool exhaustion under concurrent sessions	LOW	MEDIUM	TiDB serverless auto-scales; monitor in first week

Recommendation

GO. Proceed with the Alghanim Industries pilot as planned. The platform is certified, rehearsed, and ready. The Conglomerate Enterprise Blueprint provides a repeatable playbook for onboarding additional GCC conglomerate clients after the Alghanim pilot.

Document Index

Document	Path	Status
CR-6 Performance Report	docs/cert.cr6.performance-report.md	FINAL
Conglomerate Enterprise Blueprint	docs/pilot.conglomerate-enterprise-blueprint.md	FINAL
End-to-End Pilot Rehearsal Report	docs/pilot.rehearsal-report.md	FINAL
Active Backlog Integrity Report	docs/backlog.active-integrity-report.md	FINAL
Enterprise Certification Report (this document)	docs/cert.enterprise-final-report.md	FINAL

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